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Prova finale: Introduzione all'analisi di missioni spaziali

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1. Introduction

The primary objective of the assignment is to ascertain potential orbital transfers capable of repositioning a spacecraft from a designated initial position within one orbit to a specified final position along a distinct orbit characterized by different parameters. Each team member contributed conceptual ideas for conceivable transfers, subsequently collaboratively refining and elaborating upon them. Individual team members utilized MATLAB to formulate bespoke codes for the computation of the strategies. Subsequently, a comparative analysis was conducted to identify and rectify any discrepancies.

The outcomes were then transcribed onto an Excel spreadsheet, facilitating a comprehensive review and discussion of the data to discern optimal manoeuvres. Six overarching combinations encompassing potential manoeuvres emerged, resulting in a total of twenty possibilities, each exerting an impact on the mission in terms of temporal considerations and fuel expenditures. This report presents a singular representative possibility for each macro category, with a comprehensive compilation of all possibilities available in the appendix.

2. Initial and final orbit characterization

2.1. Initial orbit characterization

Given data	$r_{1,x}$ [km]	$r_{1,y}$ [km]	$r_{1,z}$ [km]	$v_{1,x}$ [km/s]	$v_{1,y}$ [km/s]	$v_{1,z}$ [km/s]
value	5384.9277	-7526.7861	-5058.3604	4.357	3.738	-1.265

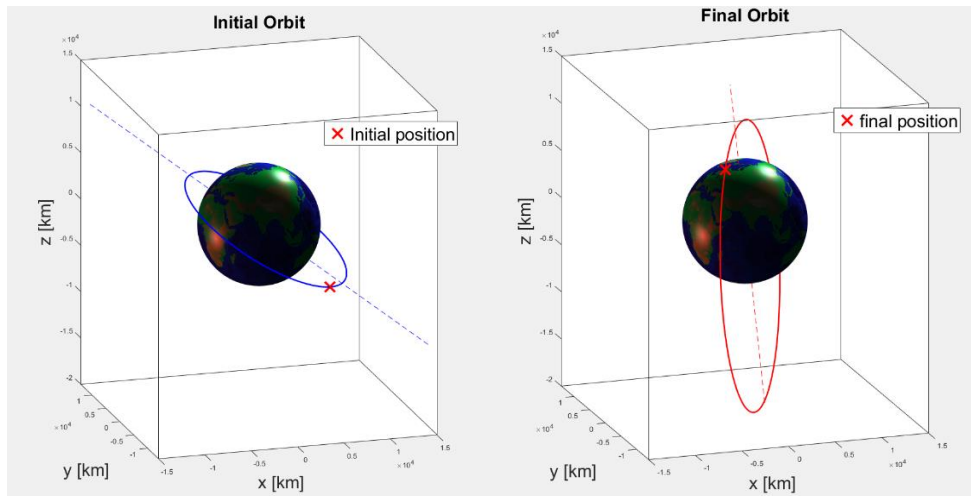
Orbital elements	a_1 [km]	e_1 [-]	i_1 [deg]	Ω_1 [deg]	ω_1 [deg]	θ_1 [deg]	
value	9714.85	0.09	31.3577	61.8257	83.5846	163.5837	
Orbital elements	h_1 [km ² /s]	p_1 [km]	$r_{p,1}$ [km]	$r_{a,1}$ [km]	$v_{p,1}$ [km/s]	$v_{a,1}$ [km/s]	T_1 [s]
value	61975.49	9636.12	8840.28	10589.43	7.0106	5.8526	9529

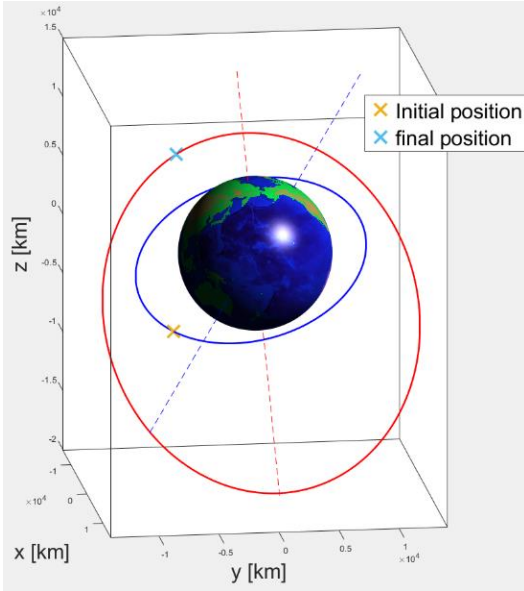
2.2. Final orbit characterization

Given data	a_2 [km]	e_2 [-]	i_2 [deg]	Ω_2 [deg]	ω_2 [deg]	θ_2 [deg]
value	15030	0.3333	85.7718	59.9887	97.9185	35.9875

Orbital elements	$r_{2,x}$ [km]	$r_{2,y}$ [km]	$r_{2,z}$ [km]	$v_{2,x}$ [km/s]	$v_{2,y}$ [km/s]	$v_{2,z}$ [km/s]
value	-4133.83	-6039.2	7560.63	-2.6124	-5.118	-4.0277

Orbital elements	h_2 [km ² /s]	p_2 [km]	$r_{p,2}$ [km]	$r_{a,2}$ [km]	$v_{p,2}$ [km/s]	$v_{a,2}$ [km/s]	T_2 [s]
value	72975.58	13360.33	10020.5	20039.5	7.2826	3.6416	18338





We observe that there exists no intersection between the two orbits. Consequently, it is imperative to execute a manoeuvre involving a minimum of two impulses. Furthermore, the orbits are situated in distinct planes, necessitating not only a modification in their spatial orientation but also adjustments in both their configuration and dimensions.

Both exhibit comparable values of Right Ascension of the Ascending Node (RAAN) and do not possess diametrically opposed values of pericentre anomaly, ω . This implies that there is no requirement for performing periapsis-periapsis or apoapsis-apoapsis shape alterations.

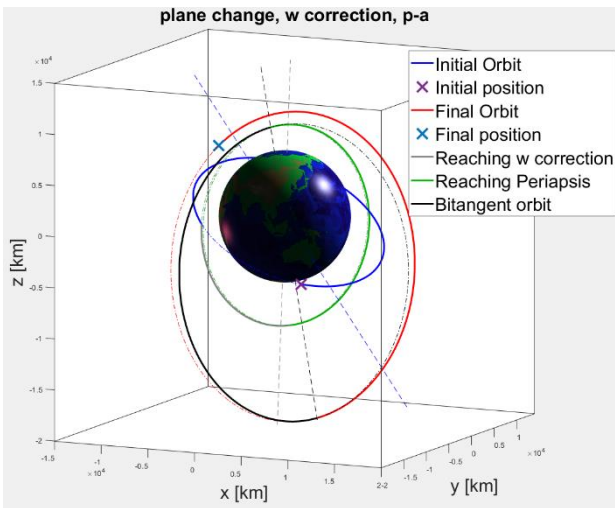
A substantial difference in inclination between the two orbits is evident, indicating that the costliest maneuver will likely be the one involving the inclination correction.

3. Transfer trajectory definition and analysis

3.1. Standard Strategy

This strategy entails the application of four distinct impulses to achieve specific orbital adjustments: the first impulse is designated for the plane change, the second for the secant burn to align with the apsidal axis, and the remaining two involve a bitangent transfer between periapsis and apoapsis to modify the orbital shape and seamlessly transition into the final orbit.

The initial phase involved determining the true anomalies for the two possible manoeuvres required to execute the plane change. The calculated values were $\theta_{\text{man},1} = 94.1629^\circ$ and $\theta_{\text{man},2} = 274.1629^\circ$. The decision was made to utilize $\theta_{\text{man},1}$, as selecting the angle with a negative cosine, while not advantageous in terms of time, resulted in a gain in Δv , because the plane change Δv is directly proportional to the transverse velocity. Subsequently, for the secant burn, there were two possible choices for the burn: 186.6283° or 6.6283° . Considering the previously determined $\theta_{\text{man},1}$, the decision was made to execute the burn at 186.6283° .



Although the Δv values were the same for both angles, this choice allowed for savings in terms of Δt .

Following this, the last two impulses were calculated after characterizing the transfer orbit for the bitangent transfer.

Once the parameters were computed, determining the Δv required at periapsis to enter the transfer orbit the necessary Δv at apoapsis to enter the final orbit became a straightforward process.

At last, Δt values were obtained from:

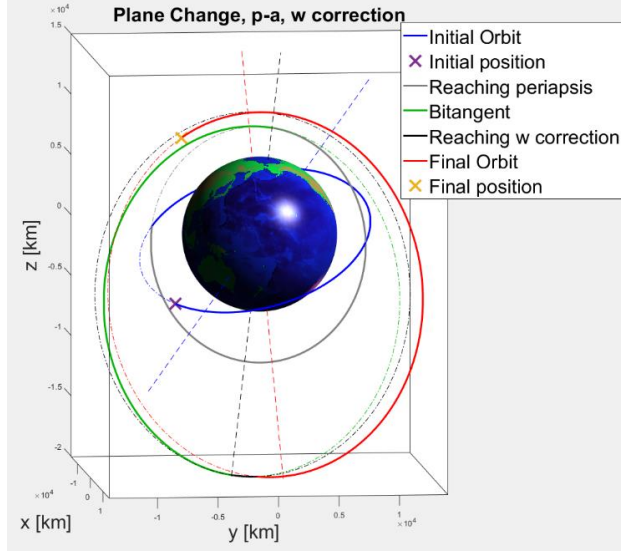
- θ_1 to the chosen $\theta_{\text{man},1}$ on the initial orbit.
- $\theta_{\text{man},1}$ to 186.6283° in the first transfer orbit.
- Time to reach the periapsis of the second transfer orbit.
- Time to reach the apoapsis of the bitangent transfer orbit.
- Time to reach θ_2 on the destination orbit.

All alternative possibilities for this transfer, involving the same manoeuvres, are comprehensively detailed in the accompanying table (Appendix, Table A).

3.2. Alternative Strategies

3.2.1 Alternative Strategy 1

This strategy revolves around changing the standard strategy's burn order: Plane change, bitangent shape change periapsis-apoapsis and secant ω correction at the true anomaly chosen as seen before. For the calculation of the first three impulses, we adopted reasoning analogous to the standard strategy outlined earlier, resulting in similar values.



The difference arises in calculating the impulse of the secant maneuver, as it is proportional to $\frac{e}{\sqrt{p}}$ and in this scenario the ratio is higher than the previous case, leading to a higher Δv .

The total time for this strategy comprises time to go from:

- the initial position to $\theta_{\text{man},1}$ (94.1629°)
- $\theta_{\text{man},1}$ to the pericenter of the first transfer orbit.
- the pericenter to the apocenter of the bitangent transfer orbit.
- the apocenter of the last transfer orbit to $\theta_{\text{secant}} = 186.6283^\circ$
- $\theta_{\text{secant},2} = 173.3717^\circ$ to θ_2 in the final orbit.

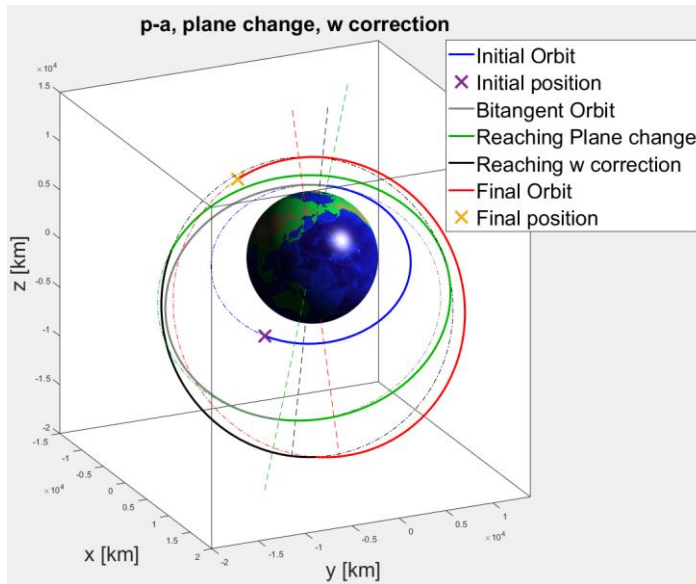
All other possible alternatives to this transfer, with the same maneuvers, are outlined in the table.

(Appendix, Table B)

3.2.2 Alternative Strategy 2

This strategy is another combination of the maneuvers involved in the standard strategy: bitangent periapsis-apoapsis shape change, plane change, secant maneuver. Once the bitangent orbit is characterized it is immediate to calculate the first two impulses. Subsequently, for the plane change, we found the maneuver true anomalies (analogous to the other strategies).

After calculating the corresponding impulse, we executed a secant maneuver at $\theta_{\text{secant}} = 186.6283^\circ$ (using the same reasoning as explained earlier).



By performing the plane change after the shape-changing burn (which retains the same impulse), a smaller impulse is required. This is attributed to the maneuver angles having an almost negligible cosine, emphasizing the influence of $\sqrt{p}$. Due to the larger semi latus rectum, it leads to a reduction in Δv .

The calculation of the total time for executing this strategy includes the time to go from:

- θ_1 to the pericenter of the initial orbit,
- p-a on the bitangent transfer orbit,
- apoapsis to $\theta_{\text{man},1}$ to execute the plane change.
- $\theta_{\text{man},1}$ to θ_{secant} to execute the secant burn.
- $\theta_{\text{secant},2} = 173.3717^\circ$ to θ_2 on the destination orbit.

All other possible alternatives to this transfer, with the same maneuvers, are detailed in the table.

(Appendix, Table C)

3.2.3 Direct Orbital Transfer from Start Point to End Point

The objective of this orbital transfer is to minimize the time required for the transfer by connecting the start and end points directly through a single transfer orbit, thus avoiding traversal of more favorable orbits. The transfer is accomplished with two burns: the first at the initial point to transition on the transfer orbit and the second at the final point to move from the transfer orbit to the destination orbit.

To initiate the analysis, the distances from the focal point to the start and end points are compared, resulting in $r_1 = 10546.89$ km, $r_2 = 10522.53$ km. While it is theoretically possible to identify an infinite number of transfer orbits passing through the specified points, this report focuses on a singular case. This case is determined by setting r_1 as the apoapsis radius of the transfer orbit (valid as greater than r_2).

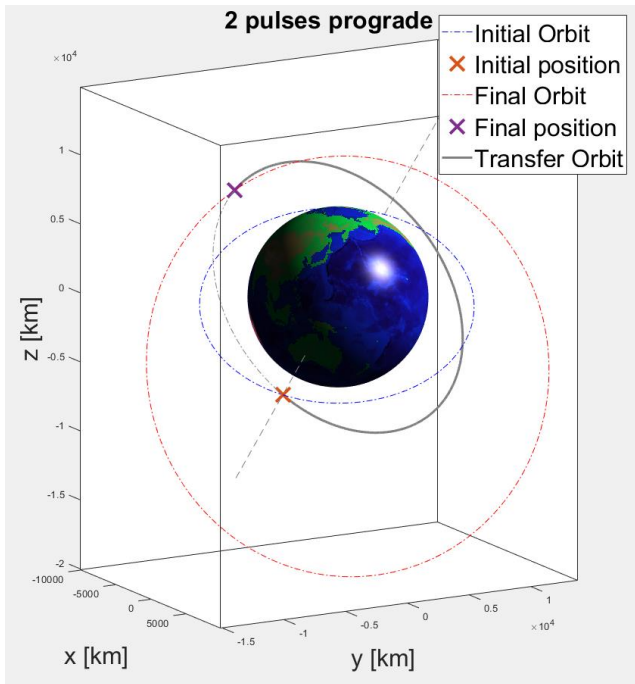
This orbit can be traversed in either a prograde (a) or retrograde (b) manner, creating two sub-cases. For this discussion, we concentrate on the prograde case; the retrograde orbit's details can be found in tables (Appendix, Table D).

Given the assumption $\vec{r}_{a,t} = \vec{r}_1$, the true anomaly θ at the initial point is π . To calculate θ at the end point, the angle between $\vec{r}_1$ and $\vec{r}_2$ is computed using the scalar product definition. As the initial point corresponds to the apoapsis ($\theta_{a,t} = \pi$), the angle at the final point on the transfer orbit ($\theta_{2,t}$) is derived by subtracting the obtained angle from π :

$$\theta_{2,t} = \pi - \cos^{-1} \left[\frac{(\vec{r}_{a,t} \cdot \vec{r}_2)}{|\vec{r}_{a,t}| |\vec{r}_2|} \right]$$

Once $\theta_{2,t}$ is determined, the values can be substituted into the generic radius formula as a function of θ , and the relationship inverted to obtain the eccentricity.

$$e_t = \frac{r_{at} - r_2}{r_{at} + r_2 \cos \theta_{2,t}}$$



As the orbit directly links the initial and final points, the direction of the vector $\hat{h}_t$ is identified by a vector product between $\vec{r}_{a,t}$ and $\vec{r}_2$, divided by their norms. Simultaneously, the direction of the velocity at the apoapsis ($\hat{v}_{at}$) is determined, while its modulus is easily obtained from the calculated eccentricity and the semi latus rectum of the transfer orbit.

$$\hat{h}_t = \frac{(\vec{r}_{a,t} \wedge \vec{r}_2)}{|\vec{r}_{a,t} \wedge \vec{r}_2|} \quad \hat{v}_{at} = \frac{(\hat{h}_t \wedge \vec{r}_{a,t})}{|\hat{h}_t \wedge \vec{r}_{a,t}|}$$

With the velocity vector at the apoapsis at disposal, $\vec{h}_t$ and $\vec{e}_t$ vectors can be determined, facilitating the full characterization of i_t , Ω_t and ω_t of the transfer orbit.

Finally, the maneuvering Δv is calculated as the sum of the two impulses. The first impulse is obtained through the vector difference between $\vec{v}_{a,t}$ at the apoapsis of the transfer orbit and $\vec{v}_1$ at the initial point. The second impulse results from the vector difference between the velocity at the arrival point ($\vec{v}_2$) and at the same point in the transfer orbit ($\vec{v}_{2,t}$).

The time required to execute the maneuver is the arc traversed from the apocentre to $\theta_{2,t}$ along the transfer orbit.

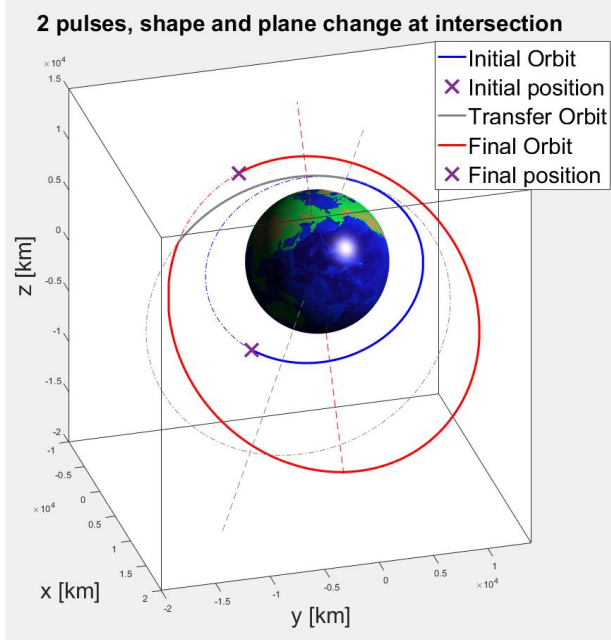
3.2.4 Two-Burns Transfer on the Initial Orbit Plane

In this scenario, a two-pulse transfer is achieved using only one transfer orbit. The objective is to ensure that the transfer orbit is coplanar with the initial orbit, intersecting the final orbit at one of the two points where they intersect the orbital plane, 94.1629° and 274.1629° as seen before.

Since the transfer orbit lies in the same plane as the initial orbit, we consider cases where its periapsis coincides with the apoapsis of the initial orbit and cases where it coincides with the periapsis of the initial orbit. In both instances, the angle θ at which the second impulse occurs is one of the values found previously, as ω_t could be the same or 180° out of phase. We will illustrate the sub-case where the first impulse occurs at the periapsis of the initial orbit and the maneuver point is at $\theta_{man,t} = 94.1629^\circ$; the other three sub-cases are detailed in tables (Appendix, Table E).

The objective is to determine $\theta_{\text{man},2}$ of the maneuver point along the arrival orbit. This is achieved by identifying the direction of the maneuvering radius ($\hat{r}_{\text{man}}$), which aligns with the radius at $\theta = \theta_{\text{man},t}$ in the initial orbit ($\vec{r}_{1,\text{man}}$). Subsequently, the angle between the periapsis of the final orbit and the derived direction is calculated, representing $\theta_{\text{man},2}$.

$$\hat{r}_{\text{man}} = \frac{\vec{r}_{1,\text{man}}}{|\vec{r}_{1,\text{man}}|} \quad \theta_{\text{man},2} = \cos^{-1} \left[\frac{(\vec{r}_{p,2} \cdot \hat{r}_{\text{man}})}{|\vec{r}_{p,2}|} \right] = 80.9064^\circ$$



With $\theta_{\text{man},2}$ known, the maneuver radius r_{man} can be calculated, aiding in the derivation of the eccentricity of the transfer orbit, a crucial parameter for its characterization.

$$e_t = \frac{r_{\text{man}} - r_{p,t}}{r_{p,t} + r_{\text{man}} \cos \theta_{\text{man},2}}$$

The first impulse involves a simple tangent transfer at periapsis from the initial orbit to the transfer orbit. The second impulse constitutes a combined maneuver, and the required Δv_2 is calculated as the vector difference between the velocity of the maneuvering point in the final orbit ($\vec{v}_{\text{man},2}$) and the velocity of the maneuvering point in the transfer orbit ($\vec{v}_{\text{man},t}$).

The total time covered is the sum of three components:

- The time to travel from the initial point to the periapsis of the first orbit.
- The time to travel from the periapsis of the transfer orbit to the maneuvering point.
- The time to travel from the maneuvering point to the end point.

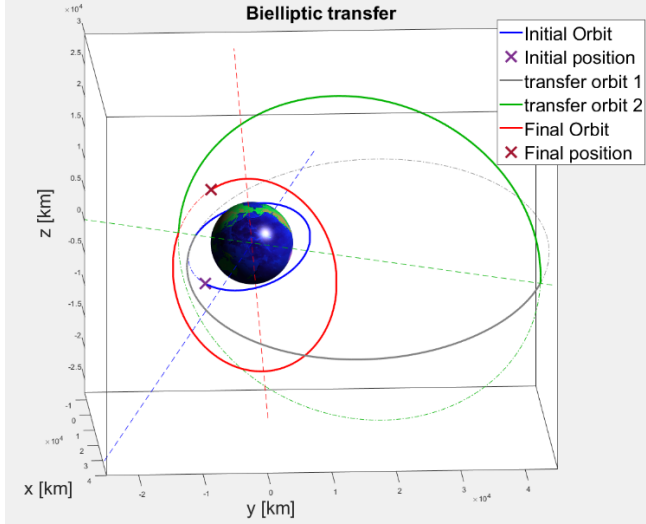
3.2.5 Bielliptic Transfer

This strategy is designed with the objective of minimizing the fuel required for changing the orbit's plane. The process involves the careful selection of a suitable pair of transfer orbits that connect the given initial and final orbits. Three impulses are necessary for this strategy: the first to move onto the first transfer orbit, the second to transition to the second transfer orbit, and the last to get onto the final orbit. In the chosen transfer orbit, the first impulse occurs at the planes' intersection when θ_{man} is 94.1629° . The relevant data concerning the first impulse at the other intersection is presented in the tables (Appendix, Table F).

During this transfer, the distance by which we move away from the focus becomes a free variable for selection. Explored various possibilities, it was determined that 50,000 km from the Earth's center is the most suitable limit for a 24-hour mission. The procedure involves calculating the true anomaly of the intersection between the planes, like the procedures discussed in the preceding cases. This information is then used to define the periapsis and apoapsis of the first transfer orbit. For example, imposing that the intersection is the periapsis of the transfer orbit. So, the characteristics of the first transfer orbit, the inclination (i), RAAN (Ω) are the same as the initial orbit, while the anomaly of the pericenter is determined by the increment of θ_{man} from the ω of the initial orbit.

Subsequently to optimize the intersection between the second transfer orbit and the final orbit, the periapsis of the second transfer orbit is set equal to the distance from the focus of the plane intersection point on the final orbit. The true anomaly of this point on the final orbit is calculated as in alternative strategy 4. The process continues with the anomaly of the pericenter which is determined with the increment of the true anomaly from the ω_2 .

$$\omega_{2,t} = \omega_2 + \theta_{\text{man},2}$$



Now the second transfer orbit is fully characterized.

Δv_1 is calculated through the vector difference of the speeds on the first transfer orbit and the initial orbit at the plane intersection. The evaluation of the Δv_2 follows using the vector difference of the speeds at the apoapsis of the transfer orbits. The Δv_3 is calculated through the vector difference between the speed on the final orbit at the previously found true anomaly and the speed at the periapsis of the second transfer orbit.

The total time includes the time to reach the plane intersection on the first orbit, half of both transfer orbits' periods, time to reach the final position from the plane intersection on the second orbit.

4. Discussion and conclusion

Considering the orbital transfers analyzed previously, we can draw some considerations regarding the time required for missions and the cost of impulses, which translates into fuel costs. The standard strategy, compared to other strategies employed, appears to be less favorable in terms of the impulse required, with comparable time requirements to other approaches.

Strategies involving a different combination of maneuvers than those in the standard strategy (Alternative 1 and 2) exhibit relatively comparable time requirements, on the order of hours. However, the first alternative proves to be more costly than the standard strategy, unlike the second, which allows for a lower fuel consumption.

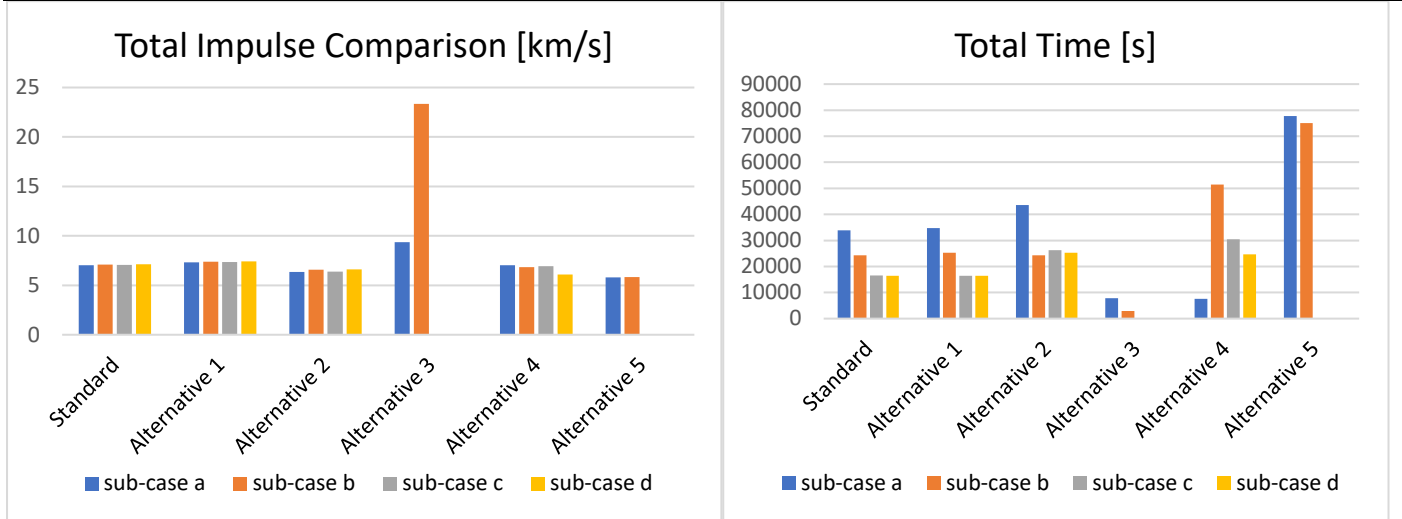
The similarity in the required times in these strategies is due to the transfer orbits obtained being identical in size and shape in the various sub-cases. Regarding velocities, the increase observed in the first alternative is due to changing the pericenter anomaly on an orbit of larger dimensions (higher $\frac{e}{\sqrt{p}}$ ratio). The decrease in the case of the second alternative is attributed to the plane change performed at greater distances from the focus.

Analyzing the direct transfer (Alternative 3), we note that, in light of a considerable time gain, the required impulse increases significantly. This is due to combined maneuvers, involving 2 plane changes, 2 shape corrections, and pericenter anomaly adjustments performed directly at the initial and final points. Among these, the sub-case with the retrograde orbit allows for the fastest transfer time but comes with a cost significantly higher than all other cases.

Studying the two-impulse transfer on the plane of the initial orbit (Alternative 4), we observe a decrease in the required velocity change compared to the previously analyzed case. This is because there is a shape change maneuver followed by a combined plane and shape change on a larger orbit. This choice proves to be favorable in terms of impulse compared to previous cases. The required time varies significantly based on the reported sub-cases.

Finally, in the bielliptic transfer (Alternative 5), it is noteworthy that all three impulses constitute combined maneuvers, albeit one of the least resource-intensive strategies, thanks to the plane change being executed at a considerable distance from the focus. In terms of time, this spacecraft repositioning plan is one of the lengthiest. It should be considered that a further reduction in required impulse could be achieved by accepting a longer duration, potentially taking days to reach the destination.

In conclusion, it can be asserted that the most favorable strategy in terms of required impulse is the bielliptic strategy (Appendix, Table F, sub-case a), while in terms of mission time, the retrograde direct transfer proves to be optimal (Appendix, Table D, sub-case b). A judicious trade-off between time and cost leads to the selection of the two-impulse strategy with a transfer orbit coplanar to the initial orbit (Appendix, Table E, sub-case d).



5. Appendix

TABLE A - B

A. Standard Strategy: Change Plane Manoeuvre + Secant Burn + Bitangent Transfer

Standard Strategy (a)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Plane Change Point t1	7501	9714.9	0.09	31.3577	61.8257	83.5846	94.1629	-
Perform Plane Change at t1	7501	9714.9	0.09	85.7718	59.9887	84.662	94.1629	5.8446
Reach Periapsis Change Point at t2	10256	9714.9	0.09	85.7718	59.9887	84.662	186.6283	-
Execute Pericenter Change at t2	10256	9714.9	0.09	85.7718	59.9887	97.9185	173.3717	0.1337
Reach Periapsis at t3	15230	9714.9	0.09	85.7718	59.9887	97.9185	0	-
Maneuver at t3	15230	14439.9	0.3878	85.7718	59.9887	97.9185	0	0.8998
Reach Apoapsis at t4	23864	14439.9	0.3878	85.7718	59.9887	97.9185	180	-
Maneuver at t4	23864	15030	0.3333	85.7718	59.9887	97.9185	180	0.152
Reach destination point at t5	33926	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								7.0301

Standard Strategy (b)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Plane Change Point t1	3280	9714.9	0.09	31.3577	61.8257	83.5846	274.1629	-
Perform Plane Change at t1	3280	9714.9	0.09	85.7718	59.9887	84.662	274.1629	5.9215
Reach Periapsis Change Point at t2	5427	9714.9	0.09	85.7718	59.9887	84.662	6.6283	-
Execute Pericenter Change at t2	5427	9714.9	0.09	85.7718	59.9887	97.9185	353.3717	0.1337
Reach Periapsis at t3	5573	9714.9	0.09	85.7718	59.9887	97.9185	0	-
Maneuver at t3	5573	14439.9	0.3878	85.7718	59.9887	97.9185	0	0.8998
Reach Apoapsis at t4	14208	14439.9	0.3878	85.7718	59.9887	97.9185	180	-
Maneuver at t4	14208	15030	0.3333	85.7718	59.9887	97.9185	180	0.152
Reach destination point at t5	24270	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								7.107

Standard Strategy (c)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Plane Change Point t1	7501	9714.9	0.09	31.3577	61.8257	83.5846	94.1629	-
Perform Plane Change at t1	7501	9714.9	0.09	85.7718	59.9887	84.662	94.1629	5.8446
Reach Periapsis Change Point at t2	10256	9714.9	0.09	85.7718	59.9887	84.662	186.6283	-
Execute Pericenter Change at t2	10256	9714.9	0.09	85.7718	59.9887	97.9185	173.3717	0.1337
Reach Apoapsis at t3	10465	9714.9	0.09	85.7718	59.9887	97.9185	180	-
Maneuver at t3	10465	10305	0.0276	85.7718	59.9887	97.9185	180	0.1974
Reach Periapsis at t4	15671	10305	0.0276	85.7718	59.9887	97.9185	0	-
Maneuver at t4	15671	15030	0.3333	85.7718	59.9887	97.9185	0	0.8891
Reach destination point at t5	16564	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								7.0648

Standard Strategy (d)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Plane Change Point t1	3280	9714.9	0.09	31.3577	61.8257	83.5846	274.1629	-
Perform Plane Change at t1	3280	9714.9	0.09	85.7718	59.9887	84.662	274.1629	5.9215
Reach Periapsis Change Point at t2	5427	9714.9	0.09	85.7718	59.9887	84.662	6.6283	-
Execute Pericenter Change at t2	5427	9714.9	0.09	85.7718	59.9887	97.9185	353.3717	0.1337
Reach Apoapsis at t3	10338	9714.9	0.09	85.7718	59.9887	97.9185	180	-
Maneuver at t3	10338	10305	0.0276	85.7718	59.9887	97.9185	180	0.1974
Reach Periapsis at t4	15544	10305	0.0276	85.7718	59.9887	97.9185	0	-
Maneuver at t4	15544	15030	0.3333	85.7718	59.9887	97.9185	0	0.8891
Reach destination point at t5	16437	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								7.1417

B. Alternative Strategy 1: Change Plane Manoeuvre + Bitangent Transfer + Secant Burn

Alternative Strategy 1 (a)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Plane Change Point t1	7501	9714.9	0.09	31.3577	61.8257	83.5846	94.1629	-
Perform Plane Change at t1	7501	9714.9	0.09	85.7718	59.9887	84.662	94.1629	5.8446
Reach Periapsis at t2	14811	9714.9	0.09	85.7718	59.9887	84.662	0	-
Maneuver at t2	14811	14439.9	0.3878	85.7718	59.9887	84.662	0	0.8998
Reach Apoapsis at t3	23445	14439.9	0.3878	85.7718	59.9887	84.662	180	-
Maneuver at t3	23445	15030	0.3333	85.7718	59.9887	84.662	180	0.152
Reach Periapsis Change Point at t4	24080	15030	0.3333	85.7718	59.9887	84.662	186.6283	-
Execute Pericenter Change at t4	24080	15030	0.3333	85.7718	59.9887	97.9185	173.3717	0.4203
Reach destination point at t5	34778	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								7.3167

Alternative Strategy 1 (b)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Plane Change Point t1	3280	9714.9	0.09	31.3577	61.8257	83.5846	274.1629	-
Perform Plane Change at t1	3280	9714.9	0.09	85.7718	59.9887	84.662	274.1629	5.9215
Reach Periapsis at t2	5282	9714.9	0.09	85.7718	59.9887	84.662	0	-
Maneuver at t2	5282	14439.9	0.3878	85.7718	59.9887	84.662	0	0.8998
Reach Apoapsis at t3	13916	14439.9	0.3878	85.7718	59.9887	84.662	180	-
Maneuver at t3	13916	15030	0.3333	85.7718	59.9887	84.662	180	0.152
Reach Periapsis Change Point at t4	14551	15030	0.3333	85.7718	59.9887	84.662	186.6283	-
Execute Pericenter Change at t4	14551	15030	0.3333	85.7718	59.9887	97.9185	173.3717	0.4203
Reach destination point at t5	25249	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								7.3936

Alternative Strategy 1 (c)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Plane Change Point t1	7501	9714.9	0.09	31.3577	61.8257	83.5846	94.1629	-
Perform Plane Change at t1	7501	9714.9	0.09	85.7718	59.9887	84.662	94.1629	5.8446
Reach Apoapsis at t2	10047	9714.9	0.09	85.7718	59.9887	84.662	180	-
Maneuver at t2	10047	10305	0.0276	85.7718	59.9887	84.662	180	0.1974
Reach Periapsis at t3	15252	10305	0.0276	85.7718	59.9887	84.662	0	-
Maneuver at t3	15252	15030	0.3333	85.7718	59.9887	84.662	0	0.8891
Reach Periapsis Change Point at t4	15411	15030	0.3333	85.7718	59.9887	84.662	6.6283	-
Execute Pericenter Change at t4	15411	15030	0.3333	85.7718	59.9887	97.9185	353.3717	0.4203
Reach destination point at t5	16464	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								7.3514

Alternative Strategy 1 (d)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Plane Change Point t1	3280	9714.9	0.09	31.3577	61.8257	83.5846	274.1629	-
Perform Plane Change at t1	3280	9714.9	0.09	85.7718	59.9887	84.662	274.1629	5.9215
Reach Apoapsis at t2	10046	9714.9	0.09	85.7718	59.9887	84.662	180	-
Maneuver at t2	10046	10305	0.0276	85.7718	59.9887	84.662	180	0.1974
Reach Periapsis at t3	15252	10305	0.0276	85.7718	59.9887	84.662	0	-
Maneuver at t3	15252	15030	0.3333	85.7718	59.9887	84.662	0	0.8891
Reach Periapsis Change Point at t4	15411	15030	0.3333	85.7718	59.9887	84.662	6.6283	-
Execute Pericenter Change at t4	15411	15030	0.3333	85.7718	59.9887	97.9185	353.3717	0.4203
Reach destination point at t5	16464	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								7.4283

TABLE C - D - E – F

C. Alternative Strategy 2: Bitangent Transfer + Change Plane Manoeuvre + Secant Burn

Alternative Strategy 2 (a)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Reach Periapsis at t1	5282	9714.9	0.09	31.3577	61.8257	83.5846	0	-
Maneuver at t1	5282	14439.9	0.3878	31.3577	61.8257	83.5846	0	0.8998
Reach Apoapsis at t2	13916	14439.9	0.3878	31.3577	61.8257	83.5846	180	-
Maneuver at t2	13916	15030	0.3333	31.3577	61.8257	83.5846	180	0.152
Plane Change Point t3	25943	15030	0.3333	31.3577	61.8257	83.5846	94.1629	-
Perform Plane Change at t3	25943	15030	0.3333	85.7718	59.9887	84.662	94.1629	4.8753
Reach Periapsis Change Point at t4	32889	15030	0.3333	85.7718	59.9887	84.662	186.6283	-
Execute Pericenter Change at t4	32889	15030	0.3333	85.7718	59.9887	97.9185	173.3717	0.4203
Reach destination point at t5	43587	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								6.3474

Alternative Strategy 2 (b)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Reach Periapsis at t1	5282	9714.9	0.09	31.3577	61.8257	83.5846	0	-
Maneuver at t1	5282	14439.9	0.3878	31.3577	61.8257	83.5846	0	0.8998
Reach Apoapsis at t2	13916	14439.9	0.3878	31.3577	61.8257	83.5846	180	-
Maneuver at t2	13916	15030	0.3333	31.3577	61.8257	83.5846	180	0.152
Plane Change Point t3	20583	15030	0.3333	31.3577	61.8257	83.5846	274.1629	-
Perform Plane Change at t3	20583	15030	0.3333	85.7718	59.9887	84.662	274.1629	5.1171
Reach Periapsis Change Point at t4	23244	15030	0.3333	85.7718	59.9887	84.662	6.6283	-
Execute Pericenter Change at t4	23244	15030	0.3333	85.7718	59.9887	97.9185	353.3717	0.4203
Reach destination point at t5	24297	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								6.5892

D. Alternative Strategy 3: Direct Orbital Transfer

Alternative Strategy 2 (c)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Reach Apoapsis at t1	517	9714.9	0.09	31.3577	61.8257	83.5846	180	-
Maneuver at t1	517	10305	0.0276	31.3577	61.8257	83.5846	180	0.1974
Reach Periapsis at t2	5722	10305	0.0276	31.3577	61.8257	83.5846	0	-
Maneuver at t2	5722	15030	0.3333	31.3577	61.8257	83.5846	0	0.8891
Plane Change Point t3	8580	15030	0.3333	31.3577	61.8257	83.5846	94.1629	-
Perform Plane Change at t3	8580	15030	0.3333	85.7718	59.9887	84.662	94.1629	4.8753
Reach Periapsis Change Point at t4	15526	15030	0.3333	85.7718	59.9887	84.662	186.6283	-
Execute Pericenter Change at t4	15526	15030	0.3333	85.7718	59.9887	97.9185	173.3717	0.4203
Reach destination point at t5	26224	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								6.3821

Alternative Strategy 2 (d)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Reach Apoapsis at t1	517	9714.9	0.09	31.3577	61.8257	83.5846	180	-
Maneuver at t1	517	10305	0.0276	31.3577	61.8257	83.5846	180	0.1974
Reach Periapsis at t2	5722	10305	0.0276	31.3577	61.8257	83.5846	0	-
Maneuver at t2	5722	15030	0.3333	31.3577	61.8257	83.5846	0	0.8891
Plane Change Point t3	21558	15030	0.3333	31.3577	61.8257	83.5846	274.1629	-
Perform Plane Change at t3	21558	15030	0.3333	85.7718	59.9887	84.662	274.1629	5.1171
Reach Periapsis Change Point at t4	24219	15030	0.3333	85.7718	59.9887	84.662	6.6283	-
Execute Pericenter Change at t4	24219	15030	0.3333	85.7718	59.9887	97.9185	353.3717	0.4203
Reach destination point at t5	25272	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								6.6239

E. Alternative Strategy 4: Two-Burns Transfer on the Initial Orbit Plane

Alternative Strategy 3 (a)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Maneuver at t0	-	10525.5	0.002	54.6381	102.759	36.0226	180	3.6692
Reach destination point at t1	7821	10525.5	0.002	54.6381	102.759	36.0226	82.2066	-
Maneuver at t1	7821	15030	0.3333	85.7718	59.9887	97.9185	35.9875	5.6999
								9.3691

Alternative Strategy 3 (b)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Maneuver at t0	-	10525.5	0.002	125.362	282.759	143.9774	180	11.4491
Reach destination point at t1	2926	10525.5	0.002	125.362	282.759	143.9774	277.7934	-
Maneuver at t1	2926	15030	0.3333	85.7718	59.9887	97.9185	35.9875	11.907
								23.3561

Alternative Strategy 4 (a)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Reach Apoapsis at t1	517	9714.9	0.09	31.3577	61.8257	83.5846	180	-
Maneuver at t1	517	15183.5	0.3026	31.3577	61.8257	263.5846	0	1.1496
Reach the Maneuver Point at t2	3597	15183.5	0.3026	31.3577	61.8257	263.5846	94.1629	-
Maneuver at t2	3597	15030	0.3333	85.7718	59.9887	97.9185	260.9064	5.8706
Reach destination point at t3	7576	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								7.0202

Alternative Strategy 4 (b)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Reach Periapsis at t1	5282	9714.9	0.09	31.3577	61.8257	83.5846	0	-
Maneuver at t1	5282	27060.8	0.6733	31.3577	61.8257	83.5846	0	1.6755
Reach the Maneuver Point at t2	47424	27060.8	0.6733	31.3577	61.8257	83.5846	274.1629	-
Maneuver at t2	47424	15030	0.3333	85.7718	59.9887	97.9185	260.9064	5.1474
Reach destination point at t3	51403	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								6.8229

F. Alternative Strategy 5: Bielliptic Transfer

Alternative Strategy 4 (c)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Reach Apoapsis at t1	517	9714.9	0.09	31.3577	61.8257	83.5846	180	-
Maneuver at t1	517	13532	0.2175	31.3577	61.8257	263.5846	0	0.9169
Reach the Maneuver Point at t2	13508	13532	0.2175	31.3577	61.8257	263.5846	274.1629	-
Maneuver at t2	13508	15030	0.3333	85.7718	59.9887	97.9185	80.9064	6.017
Reach destination point at t3	30432	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								6.9339

Alternative Strategy 4 (d)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Reach Periapsis at t1	5282	9714.9	0.09	31.3577	61.8257	83.5846	0	-
Maneuver at t1	5282	14601.3	0.3946	31.3577	61.8257	83.5846	0	0.9191
Reach the Maneuver Point at t2	7687	14601.3	0.3946	31.3577	61.8257	83.5846	94.1629	-
Maneuver at t2	7687	15030	0.3333	85.7718	59.9887	97.9185	80.9064	5.1785
Reach destination point at t3	24611	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								6.0976

Alternative Strategy 5 (a)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Initial Point t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Reach the Maneuver Point at t1	7501	9714.9	0.09	31.3577	61.8257	83.5846	94.1629	-
Maneuver at t1	7501	29849.8	0.6751	31.3577	61.8257	177.7475	0	1.9927
Reach Apoapsis at t2	33163	29849.8	0.6751	31.3577	61.8257	177.7475	180	-
Maneuver at t2	33163	31345.9	0.5951	85.7718	59.9887	178.8249	180	1.5667
Reach Periapsis at t3	60779	31345.9	0.5951	85.7718	59.9887	178.8249	0	-
Maneuver at t3	60779	15030	0.3333	85.7718	59.9887	97.9185	80.9064	2.235
Reach destination point at t4	77703	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								5.7944

Alternative Strategy 5 (b)	t [s]	a [km]	e [-]	i [deg]	Ω [deg]	ω [deg]	θ [deg]	Δv [km/s]
Punto iniziale t0	-	9714.9	0.09	31.3577	61.8257	83.5846	163.5837	-
Reach the Maneuver Point at t1	3280	9714.9	0.09	31.3577	61.8257	83.5846	274.1629	-
Maneuver at t1	3280	34786.8	0.7248	31.3577	61.8257	357.7475	0	2.0823
Reach Apoapsis at t2	35565	34786.8	0.7248	31.3577	61.8257	357.7475	180	-
Maneuver at t2	35565	37051.6	0.6194	85.7718	59.9887	358.8249	180	1.3622
Reach Periapsis at t3	71054	37051.6	0.6194	85.7718	59.9887	358.8249	0	-
Maneuver at t3	71054	15030	0.3333	85.7718	59.9887	97.9185	260.9064	2.4005
Reach destination point at t4	75033	15030	0.3333	85.7718	59.9887	97.9185	35.9875	-
								5.845